WEEK 7

STUDENT PERFORMANCE DATA ANALYSIS USING POWER BI

Data Cleaning • Data Transformation • DAX Measures

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Tool Used: Microsoft Power BI
Project: Student Performance Data Analysis Using Power BI

1. Introduction

The Student Performance Analysis project focuses on preparing, transforming, and analyzing student examination data using Microsoft Power BI. The project aims to improve data quality and organize student scores for meaningful analysis.

Power Query was used to inspect, clean, and transform the dataset, while DAX measures were created to calculate subject-wise averages, overall performance, and pass percentage.

2. Objectives

- To import and examine the Student Performance in Exams dataset.
- To identify and handle missing values, errors, and duplicate records.
- To clean and standardize categorical attributes.
- To create calculated columns for total score, average score, and performance grade.
- To classify students into Pass and Fail categories.
- To reshape subject scores using the Unpivot Columns feature.
- To create basic DAX measures for performance analysis.
- To validate the transformed dataset before visualization.

3. Tools and Technologies Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Student Performance in Exams dataset

4. Dataset Description

The dataset contains student examination information, including demographic details, parental education, lunch type, test preparation status, and examination scores.

Important attributes:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score

- Reading Score
- Writing Score

These attributes help examine student performance across different subjects and student groups.

5. Data Import and Inspection

The dataset was imported into Power BI Desktop using the Get Data option. Transform Data was selected to open Power Query Editor.

The columns, data types, rows, and categorical attributes were inspected. Missing values, errors, and duplicate records were checked to understand the quality of the original dataset.

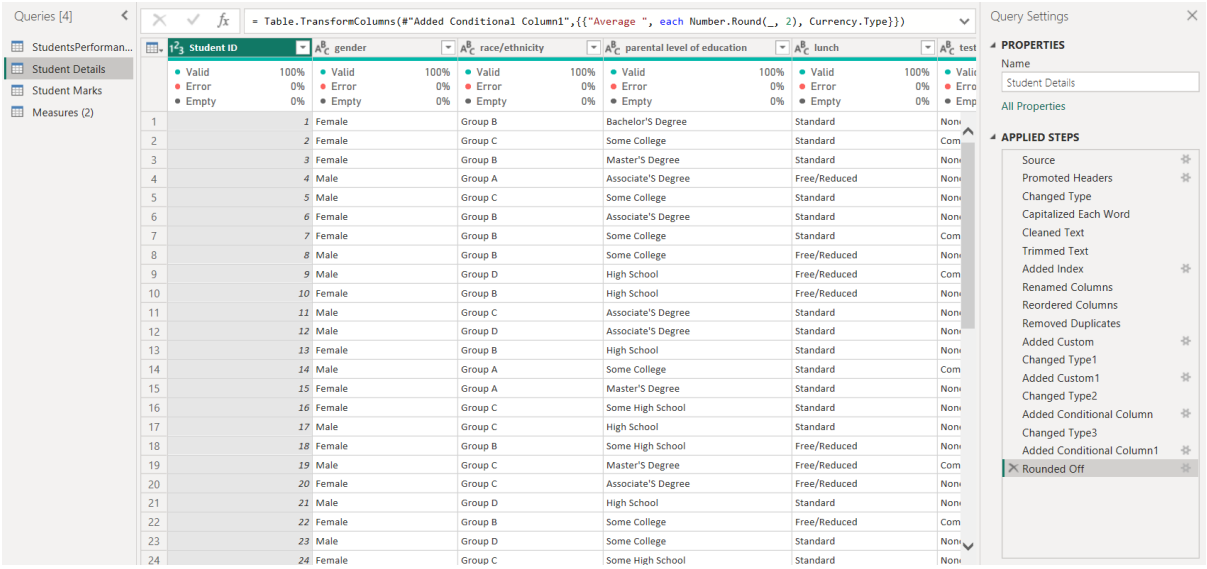
6. Data Cleaning and Transformation

The dataset was cleaned and prepared in Power Query Editor.

The following operations were performed:

- Checked for duplicate records and removed duplicates where appropriate.
- Inspected missing and invalid values.
- Verified and corrected data types.
- Standardized categorical values where necessary.
- Renamed columns for better readability.
- Reviewed demographic and examination-related attributes.

These steps improved data consistency and prepared the dataset for further analysis.



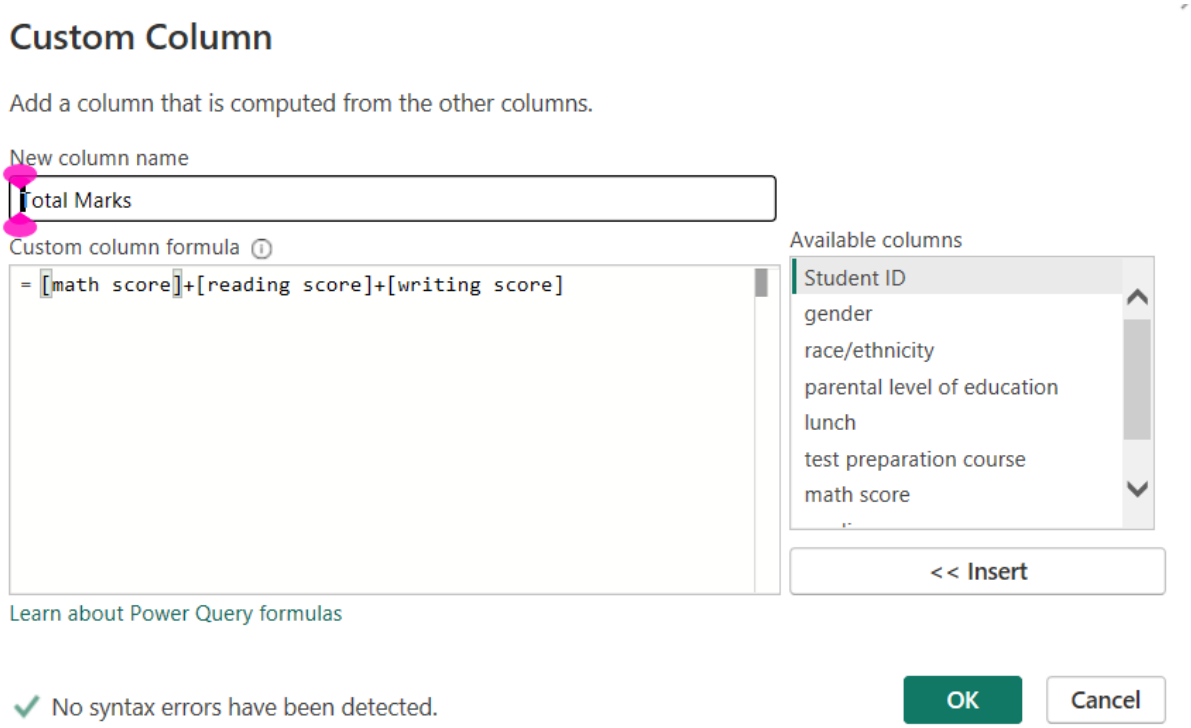
7. Calculated Columns

Additional columns were created to derive meaningful information from the examination scores.

7.1 Total Score

The total score was calculated by adding the Math, Reading, and Writing scores.

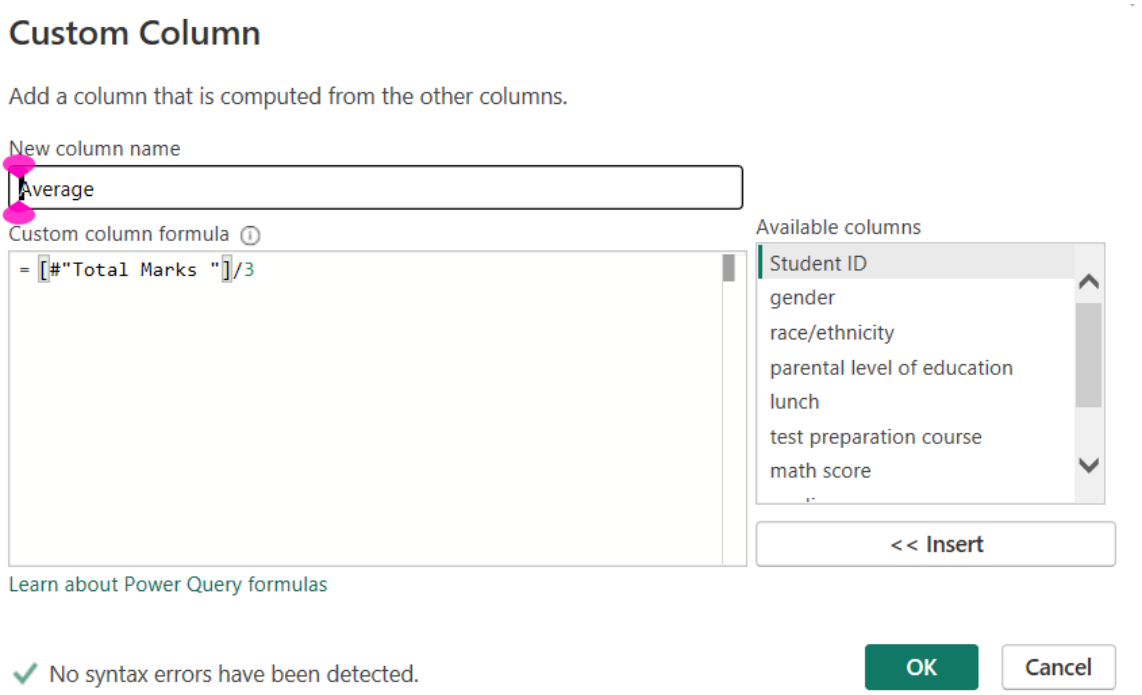
Formula: Total Score = Math Score + Reading Score + Writing Score



7.2 Average Score

The average score was calculated by dividing the total score by three.

Formula: Average Score = Total Score / 3

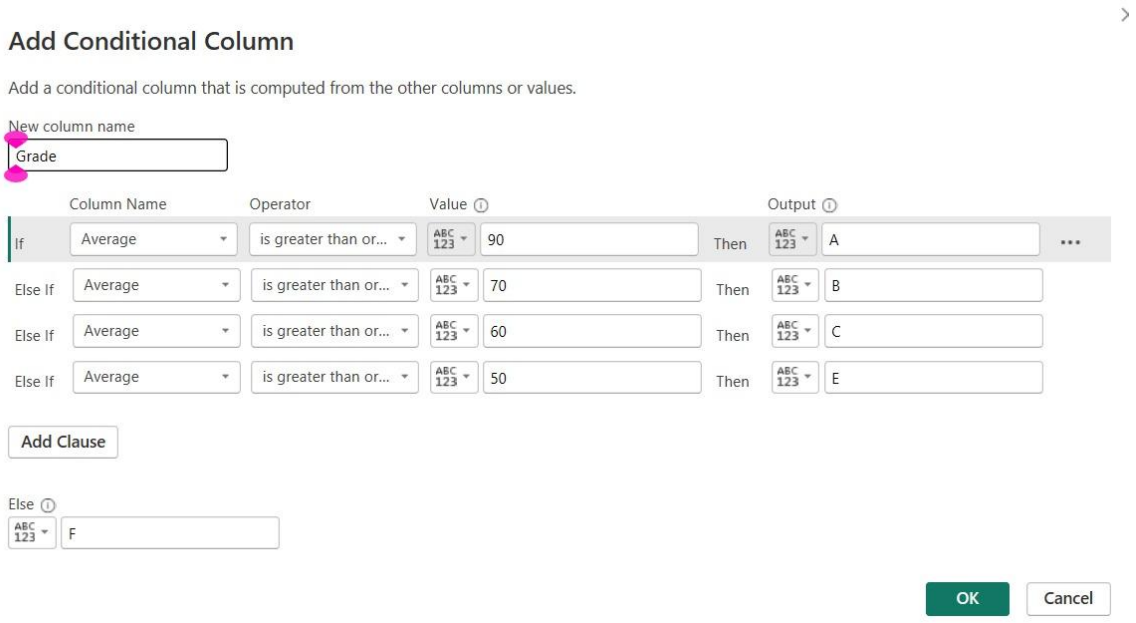


7.3 Performance Grade

A conditional classification was created to group students according to their average scores.

- Average ≥ 90 → A
- Average ≥ 70 → B
- Average ≥ 60 → C
- Average ≥ 50 → F
- Else → F

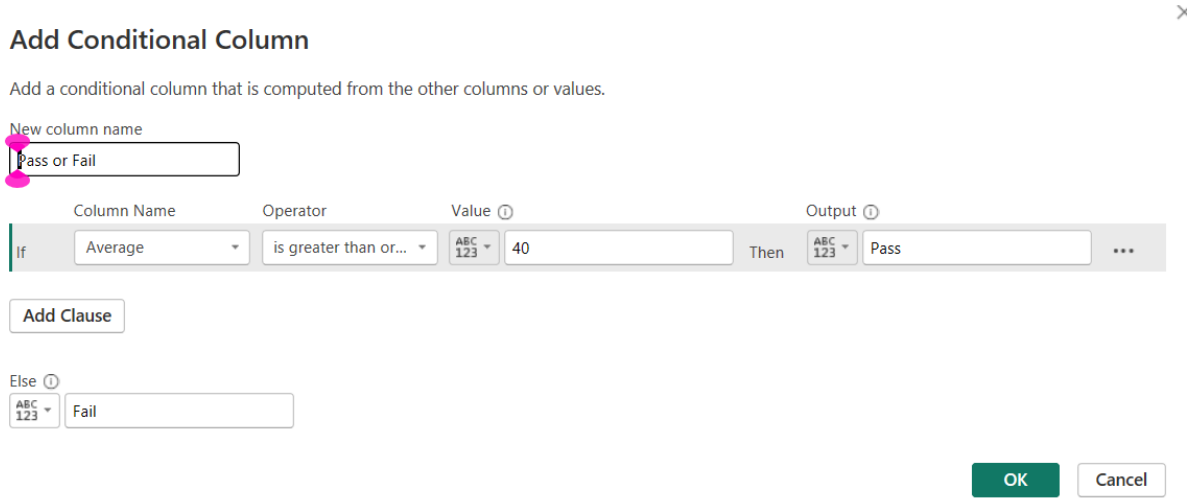
The score ranges for each grade should be defined consistently before applying the classification.



8. Pass/Fail Classification

A Conditional Column was used to classify students as Pass or Fail based on a defined passing criterion.

For example, a student may be classified as Pass if the student scores at least 40 in all three subjects; otherwise, the student is classified as Fail. This criterion is an example and should match the rule selected for the project.



9. Unpivoting Subject Scores

9.1 Before Unpivoting

Before unpivoting, Math Score, Reading Score, and Writing Score were stored in separate columns in the dataset.

Student ID	math score	reading score	writing score
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	72	72	74
2	69	90	88
3	90	95	93
4	47	57	44
5	76	78	75
6	71	83	78
7	88	95	92
8	40	43	39
9	64	64	67
10	38	60	50
11	58	54	52
12	40	52	43
13	65	81	73
14	78	72	70
15	50	53	58

9.2 Applying the Unpivot Transformation

The Math Score, Reading Score, and Writing Score columns were selected in Power Query Editor. The Unpivot Columns feature was applied to convert the subject columns into rows.

The resulting columns were renamed as Subject and Score.

Student ID		Subject		Score	
Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%
1		math score		72	
1		reading score		72	
1		writing score		74	
2		math score		69	
2		reading score		90	
2		writing score		88	
3		math score		90	
3		reading score		95	
3		writing score		93	
4		math score		47	
4		reading score		57	
4		writing score		44	
5		math score		76	
5		reading score		78	

9.3 Result of Unpivoting

This transformation converted the subject score columns into rows, making it easier to compare student performance across Math, Reading, and Writing. The transformed dataset can now support subject-wise analysis and visualizations in Power BI.

10. Data Model Preparation and Loading

The transformed dataset was reviewed to identify unnecessary columns and ensure that the data structure supported analysis.

The cleaned data was loaded into Power BI using Close & Apply. The resulting tables and transformed columns were checked to ensure that the data was ready for analysis.

Additional tables and relationships were considered where appropriate to support a structured data model.

11. DAX Measures

Basic DAX measures were prepared to calculate key performance indicators.

The measures included:

- **Average Math Score**

  Average Math Score = AVERAGE('Student Details'[math score])

- **Average Reading Score**

  Average Reading Score = AVERAGE('Student Details'[reading score])

- **Average Writing Score**

  Average Writing Score = AVERAGE('Student Details'[writing score])

- **Overall Average Score**

  Average Score = AVERAGE('Student Marks'[Score])

- **Overall Pass Percentage**

  Pass% = DIVIDE(CALCULATE(COUNT('Student Details'[Pass or Fail]),'Student Details'[Pass or Fail]="Pass"),DISTINCTCOUNT('Student Details'[Student ID]),0)

These measures help summarize student performance and provide a foundation for future Power BI dashboards.

12. Data Validation

The transformed data was validated by comparing calculated values with the original examination scores.

The following checks were considered:

- Verification of total and average scores.
- Validation of grade and Pass/Fail classifications.
- Confirmation of the unpivoted Subject and Score columns.
- Review of missing values, errors, and duplicate records.
- Verification of DAX measure results.

13. Outcome

The Student Performance dataset was prepared for analysis through data inspection, cleaning, transformation, and calculated attribute creation. Subject scores were reshaped into a structured format, and basic DAX measures were prepared to support performance evaluation.

The resulting dataset provides a foundation for comparing subject-wise scores, identifying performance patterns, and developing interactive Power BI visualizations.

14. Conclusion

This week's work provided practical experience in data preparation and transformation using Power BI. The project demonstrated how Power Query improves data quality and how calculated columns and DAX measures support analytical tasks. The cleaned and structured dataset is now suitable for further exploration and student performance reporting.